

GeoBridge: A Semantic-Anchored Multi-View Foundation Model Bridging Images and Text for Geo-Localization

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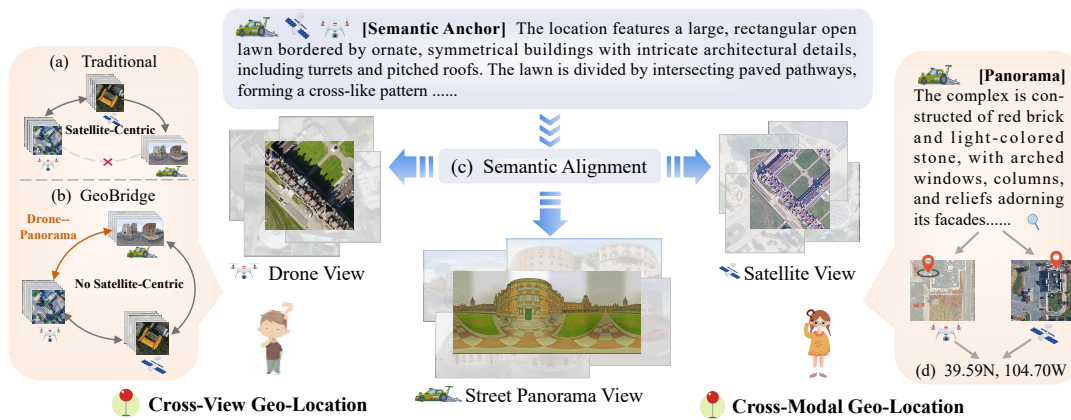


Figure 1. Schematic diagram of GeoBridge. Cross-view geo-location aims to match images with geo-referenced coordinates based on images, while cross-modal geo-location aims to match images with geo-referenced coordinates based on language descriptions. Semantic anchors are derived from multi-view images, and the localization of drone images to street view panoramic images is newly defined.

Abstract

Cross-view geo-localization infers a location by retrieving geo-tagged reference images that visually correspond to a query image. However, the traditional satellite-centric paradigm limits robustness when high-resolution or up-to-date satellite imagery is unavailable. It further underexploits complementary cues across views (e.g., drone, satellite, and street) and modalities (e.g., language and image). To address these challenges, we propose GeoBridge, a novel model that performs bidirectional matching across views and supports language-to-image retrieval. Going beyond traditional satellite-centric formulations, GeoBridge

builds on a novel semantic-anchor mechanism that bridges multi-view features through textual descriptions for robust, flexible localization. In support of this task, we construct GeoLoc, the first large-scale, cross-modal, and multi-view aligned dataset comprising over 50,000 pairs of drone, street-view panorama, and satellite images as well as their textual descriptions, collected from 36 countries, ensuring both geographic and semantic alignment. We performed broad evaluations across multiple tasks. Experiments confirm that GeoLoc pre-training markedly improves geo-localization accuracy for GeoBridge while promoting cross-domain generalization and cross-modal knowledge transfer. Code, dataset, and pretrained models will be released at <https://github.com/MiliLab/GeoBridge>.

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1. Introduction

Cross-view geo-localization seeks to infer the geographic coordinates of a query image by retrieving visually similar candidates from a database of geo-tagged reference images. By leveraging the rich spatial and semantic cues embedded in imagery, this paradigm enables high-precision localization without external annotations [8, 34]. It underpins a wide range of applications, including autonomous driving [1, 11], unmanned aerial vehicle (UAV) navigation [5, 41], geographic information retrieval [44, 47], disaster monitoring [9, 16, 32], and smart city development [24, 27, 31]. In recent years, cross-view geo-localization has seen growing interest. By learning robust feature mappings across disparate viewpoints, existing methods have substantially improved cross-domain visual correspondence and laid the groundwork for multi-source perception.

Despite this progress, core challenges remain. Owing to the strong heterogeneity between viewpoints, many methods [36, 50, 55] adopt a satellite-centric anchoring strategy for matching and retrieval (see Fig. 1a). While effective in certain conditions, this strategy under-utilizes complementary cues across views and becomes fragile when high-resolution or up-to-date satellite imagery is unavailable. An integrated framework that supports bidirectional matching across multiple views can offer alternative localization pathways and markedly improve robustness and generalization under severe viewpoint changes or missing information. Consequently, developing a unified multi-view modeling framework (see Fig. 1b) is crucial for overcoming the limitations of the “satellite-centric” localization and fully exploiting cross-view complementarity.

Meanwhile, rapid advances in large language models (LLMs) open new opportunities for geo-localization [18, 26]. Language naturally encodes rich semantics and spatial knowledge [35], as people describe places with directions, landmarks, and topological relations [33]. In practice, language-based rapid localization is especially valuable in low-bandwidth or sensor-constrained settings (*e.g.*, frontline rescue, remote areas, privacy-sensitive scenarios), where text is easier to capture and transmit than images. However, most existing studies [3, 45] overlook the complementarity between textual and vision, focusing on single-view description or retrieval. This often leads to semantic hallucinations and spatial inconsistencies. There is therefore a clear need for models that tightly couple linguistic signals with multi-view visual inputs, enhancing spatial understanding and cross-modal alignment so that locations can be inferred robustly from both descriptions and images.

To address these challenges, we propose GeoBridge, a semantic anchored, multi-view model for geo-localization (Fig. 1). During training, GeoBridge distills drone, street-view, and satellite imagery into a unified, location- and view-aware textual description that serves as a cross-modal

semantic bridge. This bridge simultaneously aligns text and each visual view to ensure cross-modal consistency, and aligns the visual views to reinforce cross-view coherence. At inference, the text branch is optional: the model can directly match any pair of drone, street-view, or satellite images, while also supporting language-to-image localization when text is available. By explicitly bridging semantic and visual spaces across views, GeoBridge moves the field beyond reliance on satellite imagery alone. It systematically enables drone-street-view matching, a long-standing gap with clear value for UAV navigation, emergency response, and low-altitude logistics. It naturally extends to air-ground multi-sensor fusion for closed-loop localization, particularly when satellite data are missing or outdated and drone imagery provides precise coordinates.

Building on this task, we construct the GeoLoc, the first large-scale, fully aligned multi-view geo-localization dataset, comprising over 50k drone images from 36 countries, each paired with a co-located Google Street View panorama and a satellite image. Leveraging GeoLoc, we pre-train GeoBridge and obtain consistent improvements in both cross-view and cross-modal geo-localization.

In summary, our contributions are as follows:

- We introduce GeoBridge, a novel semantic-anchored, flexible multi-view framework for geo-localization that performs bidirectional cross-view matching and supports cross-modal retrieval, moving beyond the traditional satellite-centric paradigm.
- We establish GeoLoc, the first large-scale, fully aligned multi-view geo-localization dataset, with more than 50,000 pairs of drone, panoramic, and satellite images from 36 countries. Each location includes a uniform textual description, and all samples are defined on geographically non-overlapping coordinates. This design supports conventional cross-view retrieval, while enabling a new and challenging drone-to-street view matching task.
- Comprehensive experiments show that, compared to existing approaches, the proposed GeoBridge method substantially improves localization accuracy across cross-view and cross-modal geo-location tasks, while further strengthening cross-domain generalization and cross-modal knowledge transfer.

2. Related Work

2.1. Geo-Location Methods

Existing cross-view geo-location methods typically follow the retrieval and matching paradigm over geo-referenced images. Early studies concentrated on retrieving satellite images given street-view queries, with robustness improved by stronger cross-domain representations [48, 55]. As low-altitude applications have expanded, drone-satellite retrieval has become increasingly important for navigation

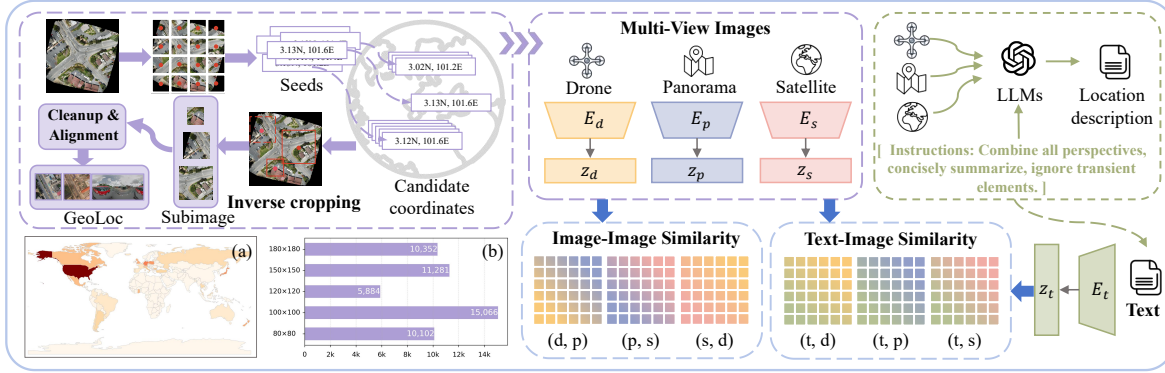


Figure 2. Overall workflow. Left: multi-view data processing for the GeoLoc dataset. Right: the GeoBridge method. (a) Global distribution of multi-view image groups (darker shades indicate higher density). (b) Counts of drone images per ground-footprint bin as a function of covered area (m²).

and positioning [4, 19, 23]. However, most approaches remain satellite-centric, making them fragile when satellite data are missing or outdated. Moreover, drone–street-view matching is still underexplored, limiting use cases such as disaster response, low-altitude logistics verification, and infrastructure inspection. In parallel, visual–language localization has begun to exhibit language-driven reasoning, but it typically relies on single-view scene descriptions, which easily lead to semantic illusions and spatial inconsistencies [3, 22, 48]. We address these gaps by shifting from a satellite-centric design to a closed-loop multi-view setting (drone–street–satellite). We further introduce a bidirectional drone–street-view matching task and incorporate position-aware semantic anchors to constrain visual representations, resulting in a more complete geo-localization process and unified cross-modal, cross-view alignment.

2.2. Geo-Location Datasets

As geo-localization research has progressed, a series of public datasets has taken shape. Early datasets such as CVUSA [39] and CVACT [20] established the standard “ground-to-satellite” setting. Later, VIGOR [53] raised the bar through dense urban sampling, panoramic views, and nearest-neighbor hard negative. Tian *et al.* [30] provided an early cross-view corpus linking street-view to bird’s-eye view imagery. For low-altitude scenarios, University-1652 [50] and SUES-200 [52] formalized “drone-to-satellite” protocols and explicitly introduced multi-height and multi-source conditions. DenseUAV [5] further contributed dense features for benchmarking drone navigation and localization. Despite these advances, existing datasets largely adhere to a two-view, satellite-centric paradigm, offering no alternative retrieval path when satellite imagery is unavailable, outdated, or latency-sensitive. In addition, their limited geographic breadth and urban morphological diversity constrain model adaptation and generaliza-

tion in complex settings. To address these gaps, we introduce the GeoLoc dataset, comprising strictly co-located drone–panorama–satellite triplets at the location level, totaling over 50,000 samples across 36 countries. While preserving standard geographic labels, GeoLoc employs multi-resolution and multi-condition acquisitions, enabling evaluation across diverse scales and conditions. This design lays a solid foundation for cross-view location research and applications, accelerates air–ground multi-sensor fusion, and enables multi-view closed-loop evaluation.

3. Method

3.1. Problem Formulation

We consider three imagery perspectives—drone (d), street-view panoramas (p), and satellite (s). Our goal is large-scale cross-view retrieval for geo-localization: given a query image from any perspective $u \in \{d, p, s\}$, retrieve co-located references from any other coordinate-bearing view $v \neq u$ and use their coordinates as the location estimate. Departing from the conventional satellite-centric paradigm, we emphasize closed-loop retrieval, where a query from any perspective can be localized via any other coordinate-bearing perspective. Moreover, it supports cross-modal retrieval initiated by a single-view textual description, enabling accurate location across diverse application scenarios.

3.2. GeoBridge

We introduce a contrastive framework GeoBridge for accurate geo-localization, which is shown in Fig. 2. It uses a unified multi-view textual description (t) as the semantic anchor to align representations across three visual perspectives: drone, street-view panoramas, and satellite. The text serves as a semantic bridge, binding view-specific visual embeddings to view-invariant scene semantics. During the training process, language supervision aligns each vi-

Table 1. Comparison of cross-view geo-localization datasets. *Platform*: A = Aerial, D = Drone, G = Street, P = Panorama, S = Satellite. *GPS-tag*: geo-coordinates provided. *Altitudes*: multiple drone heights.

Dataset	Year	Platform	Region(s)	Size(k)	GPS-tag	Altitudes
CVUSA [39]	2015	P+S	Nationwide (USA)	44.4 + 44.4	✓	—
Tian <i>et al.</i> [30]	2017	G+A	Multi-city (USA)	35.4 + 35.4	✓	—
CVACT [20]	2019	P+S	City-scale (Canberra, Australia)	128.3 + 128.3	✓	—
VIGOR [53]	2021	P+A	Multi-city (USA)	105.2 + 90.6	✓	—
University-1652 [50]	2020	D+G+S	Campus/City, multi-source	37.9 + 2.6 + 0.7	✗	✗
DenseUAV [5]	2023	D+S	City-scale (Zhejiang, China)	9.1 + 31.7	✗	✓
SUES-200 [52]	2023	D+S	Multi-city (Region)	80 + 1.6	✗	✓
GeoLoc (ours)	2025	D+P+S	Global (36 countries)	52.7 + 52.7 + 52.7	✓	✓ [‡]

[‡] GeoLoc comprises UAV imagery collected under diverse conditions, with flight altitude determined primarily by the desired ground coverage; five coverage scales are provided: 80×80 , 100×100 , 120×120 , 150×150 , and 180×180 (m²).

sual branch to a common embedding space. At inference, the model works text-free for image $\leftrightarrow$ image retrieval and naturally supports text $\rightarrow$ image search.

GeoBridge employs three view-specific visual encoders E_d , E_p , E_s , and a shared text encoder E_t . All encoders are instantiated from CLIP. Each geographic instance comprises three images captured from different perspectives. For each instance, we craft a single concise paragraph that serves as the cross-view semantic bridge. This description emphasizes stable, viewpoint-agnostic cues (*e.g.*, roads, intersections, bridges, buildings, parks, rivers, landmarks) and their relations, while omitting transient elements and viewpoint-specific phrasing.

During the training process, each example is a quadruple $(x^{(d)}, x^{(p)}, x^{(s)}, t)$. These inputs are encoded by E_d , E_p , E_s , and E_t to produce global embeddings $z_d, z_p, z_s, z_t \in \mathbb{R}^D$, $\|z\|_2 = 1$. For text, the collate function applies dynamic padding and length control via truncation: sequences exceeding the token budget T_{max} are right-truncated, and shorter ones are padded, producing batched tensors suitable for the text encoder. No additional reformatting is required, all embeddings are mapped by their encoders into a shared retrieval space.

For each view pair $(u, v) \in \{(d, p), (p, s), (s, d)\}$ and each text-to-view pair (t, v) with $v \in \{d, p, s\}$, we compute cosine similarities using cosine similarity between embeddings. A learnable temperature coefficient τ scales these similarities to control the sharpness. Considering a candidate set of size B , the score matrices are computed as:

$$S_{u,v} = z_u(z_v)^\top / \tau \in \mathbb{R}^{B \times B}, \quad (1)$$

$$S_{t,v} = z_t(z_v)^\top / \tau \in \mathbb{R}^{B \times B}. \quad (2)$$

Let y_i denote the column index of correct match for query i within the candidate set. We optimize an InfoNCE objective with a temperature-scaled softmax:

$$\mathcal{L}(S) = -\frac{1}{B} \sum_{i=1}^B \left[\log \frac{\exp(S(i, y_i))}{\sum_{j=1}^B \exp(S(i, j))} \right], \quad (3)$$

where i indexes queries and j indexes gallery items. Cross-view alignment is enforced by averaging the losses over the three image-to-image pairs:

$$\mathcal{L}_{\text{img}} = \frac{1}{3} [\mathcal{L}(S_{d,s}) + \mathcal{L}(S_{s,p}) + \mathcal{L}(S_{p,d})]. \quad (4)$$

Language-to-image alignment provides the semantic bridge, the calculation method is as follows:

$$\mathcal{L}_{\text{text}} = \frac{1}{3} \sum_{v \in \{d,p,s\}} \mathcal{L}(S_{t,v}). \quad (5)$$

The total objective aggregates the image-to-image and language-to-image terms with equal weights:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{img}} + \mathcal{L}_{\text{text}}. \quad (6)$$

This joint objective draws image-image and text-image pairs together while pushing impostors apart within a shared embedding space. At inference, we encode image or text query and retrieve by nearest neighbors in that space. For language-only queries, to avoid exposing sensitive details, the description is restricted to a single scene and retrieval is limited to other views within the same location class.

4. GeoLoc Construction and Preprocessing

To support our GeoBridge model, we curate GeoLoc, a cross-view dataset with precise geographic and semantic alignment. The process, as shown in Fig. 2, comprises four stages: (1) drone images acquisition and seed generation, (2) cross-source harvesting with inverse cropping, (3) duplicate suppression and cleanup, and (4) multi-view alignment. We detail the operations below.

Drone images acquisition and seed generation.

We collect globally sourced drone images from OpenAerialMap¹ with geo-referencing metadata (projection

¹<https://map.openaerialmap.org/>

and coordinates). For each large drone image, we slide a fixed 80×80 -pixel window to estimate the ground footprint, recording the pixel center (x, y) together with the latitude and longitude (φ, λ) . These centers serve as seeds for cross-source retrieval and alignment.

Cross-source harvesting with inverse cropping. For each seed, we query Google Street View² (GSV) to obtain candidate panoramic coordinates and retain only those whose locations fall within the coverage of drone image. Using the GSV location as an anchor, we inverse crop the original drone image according to the ground footprint to obtain a co-located drone subimage. To improve robustness to scale and viewpoint, we generate multi-scale crops whose approximate ground footprints are 80×80 , 100×100 , 120×120 , 150×150 , and 180×180 m², with matched output resolutions, yielding multi-scale drone subimages.

Duplicate suppression and cleanup. We perform an initial screening of the drone subimages. If two drone subimages have more than 50% ground-coverage overlap, or if their latitude and longitude are identical, they are treated as duplicates and only one is retained. Subimages in which more than 1% of the area consists entirely of black or entirely of white pixels (*e.g.*, sensor-strip edges) are discarded.

To further remove low-quality content, we employ a cascade of three lightweight quality gates:

- **BH-Gate (Blur-Haze):** measures global texture using Laplacian variance and pixel variance, eliminating blurry sub-images such as those with motion blur, distant scenes under heavy fog, and heavily compressed textures.
- **C-Gate (Global Contrast):** eliminates images with weak edges based on global contrast, filtering drone subimages that are slightly out of focus, over- or under-exposed.
- **UN-Gate (Uniformity-Noise):** uses entropy as the primary criterion, combined with variance range and ratios of saturated pixels, to distinguish uniform but informationless pseudo-textures from noise patterns, eliminating images lacking strong features such as large expanses of blue sky, open water, deserts, and grasslands.

In practice, this cascade yields a pool of structurally rich, high-contrast, and semantically useful crops.

Tri-view alignment. For each quality-screened drone subimage, we used its seed coordinates to retrieve Google Street View panoramas. This API returns static, non-interactive panoramic imagery. For each sampled location, we requested images using geographic coordinates together with camera parameters. Our use of the API follows service-specific terms^{3 4}. We then cropped Google Satellite⁵ tiles to align with the co-located region covered

²<https://www.google.com/streetview/>

³<https://developers.google.com/maps/documentation/streetview/policies>

⁴<https://cloud.google.com/maps-platform/terms/maps-service-terms/index-20240515>

⁵<https://www.google.com/earth/>

Table 2. Comparison on the University-1652 [50] dataset; best results are in bold, second-best are underlined.

Method	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
SAIG-D [56]	78.85	81.62	86.45	78.48
DWDR [37]	86.41	88.41	91.30	86.02
MBF [51]	89.05	90.61	92.15	84.45
MCCG [28]	89.64	91.32	94.30	89.39
SeGCN [21]	89.18	90.89	94.29	89.65
CCR [7]	92.54	93.78	95.15	91.80
MEAN [2]	93.55	94.53	96.01	92.08
DAC [42]	<u>94.67</u>	<u>95.50</u>	<u>96.43</u>	<u>93.79</u>
Sample4Geo [6]	92.65	93.81	95.14	91.39
GeoBridge (ours)	95.82	97.77	97.14	95.05

Table 3. Comparison on the SUES-200 [52] dataset; best results are in bold, second-best are underlined.

Drone to Satellite								
Method	150m		200m		250m		300m	
	R@1	AP	R@1	AP	R@1	AP	R@1	AP
MBF [51]	85.62	88.21	87.43	90.02	90.65	92.53	92.12	93.63
MCCG [28]	82.22	85.47	89.38	91.41	93.82	95.04	95.07	96.20
CCR [7]	87.08	89.55	93.57	94.90	95.42	96.28	96.82	97.39
SeGCN [21]	90.80	92.32	91.93	93.41	92.53	93.90	93.33	94.61
CGSI [29]	95.95	96.80	97.72	98.15	97.60	98.03	97.83	98.23
GeoBridge (ours)	<u>95.68</u>	97.49	97.88	98.89	97.85	98.84	<u>97.63</u>	98.62
Satellite to Drone								
Method	150m		200m		250m		300m	
	R@1	AP	R@1	AP	R@1	AP	R@1	AP
MBF [51]	88.75	84.74	91.25	89.95	93.75	90.65	96.25	91.60
MCCG [28]	93.75	89.72	93.75	92.21	96.25	96.14	98.75	96.64
CCR [7]	92.50	88.54	97.50	95.22	97.50	97.10	97.50	97.49
SeGCN [21]	93.75	92.45	95.00	93.65	96.25	94.39	97.50	94.55
CGSI [29]	<u>97.50</u>	<u>96.22</u>	<u>98.75</u>	<u>97.62</u>	<u>98.75</u>	98.01	98.75	<u>97.92</u>
GeoBridge (ours)	98.99	97.98	98.98	97.98	99.01	<u>97.98</u>	98.75	97.95

by the subimage. This produces yields 52,679 geographically co-located drone-street panorama-satellite triplets spanning 36 countries. We designate 5,351 triplets from non-overlapping cities as a held-out evaluation set. Dataset statistics are summarized in Table 1. Notably, our method remains effective on publicly datasets, indicating that its performance is not tied to any specific proprietary data source. Further details are in the supplementary materials.

5. Experiment

5.1. Experimental Setup

Implementation Details. We adopt the publicly released CLIP-L/14 backbone [25] and use ChatGPT-4o [14] to produce concise textual descriptions for images. Further details are provided in the supplementary materials. Models are trained end-to-end with Adam (base learning rate 1×10^{-5}) under a cosine decay schedule, using a batch size of 32 for 200 epochs on eight NVIDIA A800 GPUs. Input images are resized to 224×224 . During pre-training, we optimize

Table 4. Results on the CVUSA [39] and VIGOR [53] datasets; best results are shown in bold, second-best are underlined.

Method	CVUSA		VIGOR-Same		VIGOR-Cross	
	R@1	R@1%	R@1	Hit	R@1	Hit
TransGeo [54]	94.08	99.77	61.48	73.09	18.99	21.21
FRGeo [49]	97.06	99.85	71.26	82.41	37.54	40.66
SAIG-D [56]	96.08	99.86	65.23	74.11	33.05	36.71
VimGeo [13]	96.19	99.52	55.24	57.43	19.31	20.72
Sample4Geo [6]	98.68	<u>99.87</u>	77.86	89.82	61.70	69.87
Panorama-BEV [46]	98.71	99.70	<u>82.18</u>	-	<u>72.19</u>	-
AuxGeo [43]	<u>98.80</u>	99.85	80.34	93.78	63.94	<u>76.25</u>
GeoBridge (ours)	99.14	99.98	85.82	<u>92.24</u>	73.87	81.51

all parameters end-to-end; for evaluation on public benchmarks, we fine-tune only the last three layers.

Evaluation Metrics. To evaluate geo-location performance across viewpoints and modalities, we use Top-k recall ($R@k$) and mean average precision (AP) as primary metrics. $R@k$ measures whether the ground-truth reference appears within the top k retrieved results, while AP summarizes overall ranking quality across the full list. We also include the task-specific metrics as needed. For street-satellite-view retrieval, we additionally report $R@1\%$ and Hit metrics. Here, $R@1\%$ indicates whether the ground-truth sample lies within the top 1% of the gallery images, and Hit indicates that the first retrieved reference image covers the query image (including the ground-truth). For cross-modal geo-location task, we complement $R@k$ with the location recall $L@k$, which refers to the proportion of retrieved results where the distance to the actual location is below a specified threshold. When geographic coordinates are unavailable, we report $R@k$ only.

5.2. Cross-View Geo-Location Performance

For the drone-satellite matching task, we systematically evaluate GeoBridge on University-1652 [50] and SUES-200 [52] datasets, with results summarized in Table 2, 3. It is straightforward to verify that GeoBridge consistently surpasses existing methods in both robustness and overall accuracy. Among the baselines, DAC achieves strong results which adopts contrastive learning, however, its reliance on direct image-to-image alignment limits its ability to model cross-view semantic consistency and resolve spatial ambiguities. In contrast, GeoBridge introduces a semantic-anchor mechanism to align multi-view imagery within a shared semantic space, yielding simultaneous gains in $R@1$ and AP for bidirectional retrieval on University-1652 and improving overall ranking quality. These improvements are likewise evident on SUES-200 dataset. GeoBridge maintains a leading advantage across retrieval directions and altitude settings. Moreover, because the anchor operates mainly during training, incurring no additional online overhead. Taken together, these results demonstrate that GeoBridge substantially improves the accuracy and generaliza-

Table 5. Comparison on the GeoLoc dataset (D for Drone, P for Street-View Panorama, and S for Satellite). The best results are shown in bold, the second-best results are underlined.

Method	D2S		S2D		D2P		P2D	
	R@1	AP	R@1	AP	R@1	AP	R@1	AP
Sample4Geo [6]	<u>27.27</u>	<u>39.69</u>	<u>28.70</u>	<u>40.32</u>	<u>29.51</u>	<u>31.17</u>	15.56	<u>29.68</u>
MEAN [2]	21.52	27.08	21.38	26.97	13.08	17.76	1.87	7.74
DAC [42]	6.19	8.41	13.91	15.16	13.74	15.16	<u>19.34</u>	23.01
CAMP [40]	19.60	24.39	14.88	18.75	11.31	12.34	11.46	19.16
MCCG [28]	12.23	13.22	14.73	17.70	15.51	19.11	12.90	15.75
CCR [7]	12.91	15.56	13.89	16.04	10.91	12.34	10.81	14.77
GeoBridge (ours)	45.05	49.05	44.81	48.76	41.22	43.54	41.15	43.41
Method	P2S		S2P		D2P		P2D	
	R@1	AP	R@1	AP	R@1	AP	R@1	AP
panorama-BEV [46]	10.21	12.70	18.70	20.56	16.32	17.11	16.34	18.36
Sample4Geo [6]	16.82	13.64	<u>18.87</u>	17.78	<u>29.51</u>	<u>31.17</u>	15.56	<u>29.68</u>
AuxGeo [43]	13.74	17.09	15.61	<u>22.96</u>	9.34	12.69	14.70	17.40
FRGeo [49]	<u>17.13</u>	18.70	14.35	19.69	13.69	19.34	14.38	15.00
HC-Net [38]	14.17	15.05	11.79	12.03	12.83	13.31	<u>18.79</u>	18.93
TransGeo [54]	11.21	<u>23.77</u>	13.74	13.43	18.70	23.41	17.48	23.26
GeoBridge (ours)	38.87	42.10	39.20	41.96	41.22	43.54	41.15	43.41

tion of cross-view geo-location.

For street-to-satellite cross-view localization, we evaluate GeoBridge on CVUSA [39] and VIGOR [53] datasets using the standard settings of each dataset. The results summarized in Table 4. We report $R@1$, $R@1\%$, and Hit metrics. It is obvious that GeoBridge delivers clear gains on both datasets. On CVUSA, it achieves higher $R@1$ and $R@1\%$, reflecting stronger top-1 accuracy and overall recall. On VIGOR, we present results on both the Same and Cross splits. Although GeoBridge ranks second in Hit under the Same setting, its other metrics substantially exceed the second-best method. This indicates better cross-scene generalization and overall ranking quality. Overall, GeoBridge improves accuracy and robustness in street-to-satellite retrieval and demonstrates superior generalization under stringent cross-domain conditions.

To further validate the effectiveness of our proposed cross-view geo-location method, we conducted a comprehensive comparison against representative baselines on traditional view pairs (Drone-to-Satellite, Satellite-to-Drone, Street-to-Satellite, and Satellite-to-Street) on the GeoLoc dataset. We also extend these baselines to the newly introduced cross-view geo-location tasks (Drone-to-Street, and Street-to-Drone). Results are summarized in Table 5. While several prior methods excel on their preferred view pairings, their performance typically degrades when transferred to the new cross-view settings, reflecting difficulty in exploiting the multi-scale, semantically rich characteristics of GeoLoc. In contrast, GeoBridge delivers substantial and well-balanced gains across all directions, demonstrating robust retrieval under both scale and viewpoint shifts. Crucially, the improvements manifest not only in $R@1$ but also in consistently higher AP , indicating better ordering of both easy and hard inference throughout the candidate list. These results suggests that rather than relying on the inductive

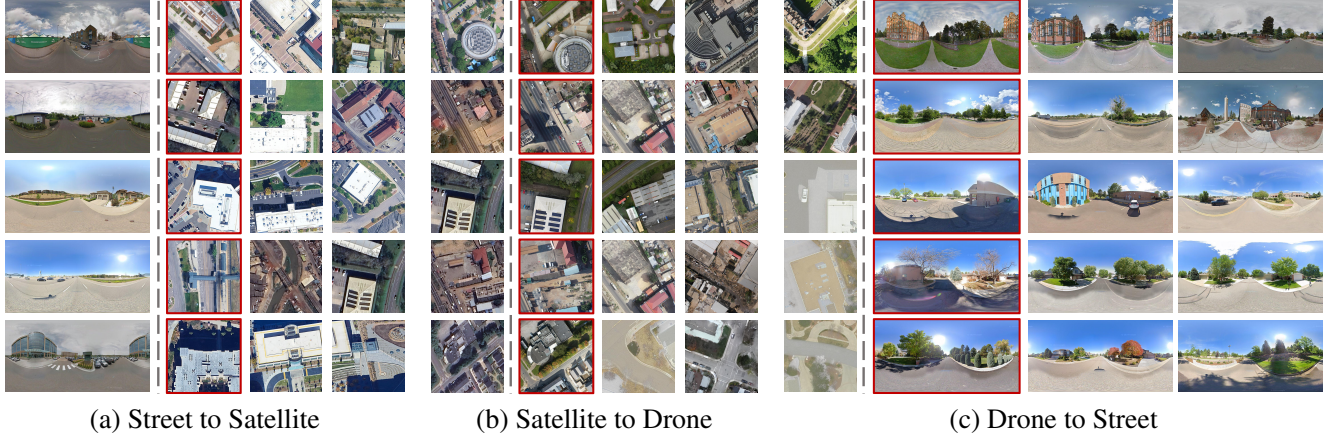


Figure 3. Qualitative image retrieval results on the GeoLoc dataset. The red boxes indicate the true-matched images.

Table 6. Comparison results on the RSIEval [12] dataset. The best results are shown in bold, the second-best results are underlined.

Method	R@1	R@5	R@10
ViLT [15]	2.00	16.00	38.00
BLIP-B [17]	0.00	1.00	2.00
EVA2-CLIP-B/16 [10]	15.00	38.00	54.00
EVA2-CLIP-L/14 [10]	19.00	43.00	55.00
CLIP-L/14 [25]	<u>25.00</u>	<u>60.00</u>	<u>74.00</u>
CLIP-B/16 [25]	18.00	51.00	61.00
CrossText2Loc[48]	12.00	36.00	67.00
GeoBridge (ours)	29.00	71.00	88.00

bias of any single view, GeoBridge effectively bridges aerial and ground-level semantics by unifying multi-view representations within a shared semantic space via semantic anchors, thereby yielding higher-quality global rankings and stronger cross-domain generalization.

Qualitative retrieval results on the GeoLoc dataset are shown in Fig. 3, covering three tasks: Street-to-Satellite, Satellite-to-Drone, and Drone-to-Street. For each query image, we display the top-3 retrieved images and highlight the ground-truth match in red. As illustrated, GeoBridge reliably retrieves the most similar candidates from a large gallery by accurately leveraging salient landmarks and structural cues. Benefiting from semantic anchors, GeoBridge maintains strong discriminative power and visual consistency across scales and resolutions, enabling robust selection of true matches from candidate lists populated with visually similar distractors. Overall, systematic experiments on GeoLoc confirm GeoBridge’s robustness and scalability across tasks, retrieval directions, and modalities, yielding a more accurate and broadly generalizable solution for cross-view geo-location. Additional visualizations are provided in the supplementary material.



Figure 4. Qualitative results for cross-modal geo-location. Using street view descriptions to match drone perspectives, the top three results are reported; red boxes indicate correct matches.

5.3. Cross-Modal Geo-Location Performance

To further assess the effectiveness of GeoBridge for cross-modal retrieval, we evaluate it on the publicly available remote sensing annotation dataset RSIEval [12], and the results are shown in Table 6. This dataset contains 100 image-text pairs. Under identical evaluation settings, GeoBridge achieves the best performance.

We further validate GeoBridge on cross-modal geo-location employing the GeoLoc dataset. Text descriptions from single-view images are used to perform cross-view retrieval (e.g., retrieving satellite or drone images from street-view text), with comparative results against representative methods reported in Table 7. Overall, GeoBridge delivers state-of-the-art performance across most text-to-image retrieval directions. CrossText2Loc, which relies on descrip-

Table 7. Comparison of multi-modal geo-location methods on GeoLoc dataset. Descriptions base on the single view; comparative results include two additional viewpoints. Best results are in bold; second-best are underlined.

Method	Street Description				Satellite Description				Drone Description			
	Satellite Image		Drone Image		Street Image		Drone Image		Street Image		Satellite Image	
	R@1	L@50	R@1	L@50	R@1	L@50	R@1	L@50	R@1	L@50	R@1	L@50
ViLT [15]	1.40	6.41	1.05	5.72	6.3	13.21	1.05	5.96	6.20	13.49	1.16	5.76
BLIP-B [17]	0.84	4.6	0.78	5.42	0.93	5.06	0.73	5.59	0.82	5.14	0.95	5.64
EVA2-CLIP-B/16 [10]	1.01	5.28	0.92	4.30	1.21	5.74	0.90	4.26	1.18	5.63	0.99	5.23
EVA2-CLIP-L/14 [10]	1.16	4.97	1.83	6.95	1.23	5.57	1.23	5.42	1.21	5.55	1.18	4.88
CLIP-L/14 [25]	<u>2.71</u>	7.81	2.02	7.96	12.20	20.71	1.89	7.21	9.18	17.38	<u>2.88</u>	7.83
CLIP-B/16 [25]	1.91	7.49	1.79	<u>8.04</u>	6.91	14.93	1.85	7.36	6.54	13.32	1.72	6.99
CrossText2Loc[48]	2.62	<u>7.76</u>	<u>2.15</u>	7.68	<u>12.91</u>	<u>22.37</u>	<u>2.35</u>	<u>8.04</u>	<u>9.25</u>	<u>17.83</u>	2.11	7.4
GeoBridge(ours)	5.68	7.61	6.10	8.76	21.58	25.53	2.99	8.85	20.96	24.73	2.93	<u>7.66</u>

tions synthesized from single-view imagery, is competitive in long-text geo-location. However, its accuracy diminishes on GeoLoc which features multi-scale scenes, complex terrain and cross-view interference. Under these conditions, precisely aligning key landmarks and spatial relationships is inherently difficult, leading to degraded performance on cross-modal retrieval tasks. By contrast, GeoBridge maps heterogeneous viewpoints into a shared semantic space via multi-view semantic anchors, thereby bridging multi-view images and textual descriptions. This design strengthens cross-scale and cross-view alignment of landmarks and structural cues, yielding more consistent ranking metrics (e.g., $R@1$, $L@50$) and maintaining a robust lead across diverse viewpoint combinations and retrieval directions.

Figure 4 shows the visualization results of retrieving drone images using short descriptions generated from street view panoramas. GeoBridge accurately captures key semantics in the descriptions (such as landmark function words, building morphology, and salient structures) and locates the corresponding regions in candidate drone images, significantly improving the ranking of correct samples. Therefore, under the cross-modal setting of single-view text query and cross-view image retrieval, GeoBridge combines accuracy and generalization, providing a more efficient and interpretable solution for cross-view geo-location. More details will be provided in the supplementary materials.

5.4. Ablation Experiment

To assess the effectiveness of semantic anchors, we conduct an ablation study on alignment strategies. The $R@1$ results are reported in Table 8. Image-only denotes image-image alignment, Text-only denotes text-image alignment, and GeoBridge fuses both while unifying multi-view representations in a shared semantic space via semantic anchors. Across all retrieval tasks, the fusion strategy attains the highest $R@1$. With image-only alignment, robust correspondences between street panoramas and markedly

Table 8. Ablation $R@1$ results for alignment strategies on GeoLoc dataset. (D for Drone, P for Street-View Panorama, and S for Satellite.) The best results are shown in bold, the second-best results are underlined.

Method	D2S	S2D	P2S	S2P	D2P	P2D
Image-only	38.20	34.63	6.43	6.95	7.16	4.90
Text-only	<u>42.83</u>	<u>42.83</u>	<u>35.40</u>	<u>36.40</u>	<u>39.00</u>	<u>38.63</u>
GeoBridge	45.06	44.81	38.87	39.21	41.23	41.15

different viewpoints (e.g., satellite and drone) are difficult to establish, yielding the weakest overall performance. In contrast, text-only alignment benefits from descriptions that capture key semantic cues across all three views, and thus ranks second overall. These results demonstrate that using location representations from multi-views as semantic anchors substantially narrows the semantic gap between ground and overhead views, improving the accuracy and robustness of cross-view retrieval.

6. Conclusion

In this work, we introduce GeoBridge, a framework for cross-view and cross-modal geo-location that performs bidirectional matching across views and supports language to image retrieval. Central to GeoBridge is a semantic anchor mechanism that aligns multi-view features through textual descriptions, enabling robust and flexible localization beyond traditional satellite centric formulations. To support this setting, we construct GeoLoc, a large scale, fully aligned dataset with more than 50,000 pairs of drone, street view panorama, and satellite images and their textual descriptions, ensuring both geographic and semantic alignment. Across a broad suite of evaluations, pre-training on GeoLoc markedly improves the localization accuracy of GeoBridge and promotes stronger cross domain generalization and cross modal knowledge transfer.

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